MIL-B-15382C(SHIPS)
19 May 1967
SUPERSEDING
MIL-B-15382B
25 March 1957, and
MIL-N-15721A
28 December 1951
(See 6.2)

MILITARY SPECIFICATION

BOLT, FIREBRICK ANCHOR

1. SCOPE

1.1 Scope. - This specification covers anchor bolts for anchoring firebricks in oil-fired boilers of Naval vessels.

1.2 Classification. - Anchor bolts shall be of the following classes and types, as specified (see 6.1):

Class A - 25 percent chromium - 20 percent nickel corrosion resistant steel.
Class B - Nickel-chromium-iron-alloy steel.
Type a - Straight end.
Type b - Hook end.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-H-3982 - Hardware (Fasteners and Related Items); Packaging and Packing for Shipment and Storage of.

FSC 5306

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STANDARDS

FEDERAL
FED-STD-151 - Metals; Test Methods.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A-276 - Hot-Rolled and Cold-Finished Corrosion-Resisting Steel Bars.
ASTM B-166 - Nickel-Chromium-Iron Alloy Rod and Bar.

(Copies may be obtained from the American Society For Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. - Anchor bolts shall be made of material conforming to ASTM A-276, type 310 for class A bolts, and ASTM B-166 for class B bolts.

3.2 Type a, straight end. - The anchor bolts shall be of the shape and dimensions shown on figure 1. The length of the shank shall be as specified (see 6.1).

3.3 Type b, hook-end. - The anchor bolts shall be of the shape and dimensions shown on figure 2. The axis of the bent portion of the shank shall be parallel to a diagonal of the head, as shown on figure 2. The length of the shank shall be as specified (see 6.1).

3.4 Dimension tolerances. - Dimensions of bolts shall be within the tolerance limits specified in table I.

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Table I - Dimension tolerances

Item	Dimension	Tolerance
	Inches	Inch
Head width	1-1/4 square	± 1/16
Head thickness	1/4	+0.06, -0.01
Shank diameter	3/8	± 0.01
Bolt length	as specified	± 1/16
Length of hook end, type b only	as specified	± 1/16

3.5 Heads. - Unless otherwise specified (see 6.1), anchor bolt heads shall be forged, resistance (flash) welded, or screwed at the option of the contractor.

3.5.1 Forged. - The forged bolts shall have a fillet under the head of not less than 1/8 nor more than 1/4 inch radius.

3.5.2 Resistance (flash) welded. - The bolts shall have heads securely flash-welded to the shank, and the weld shall withstand the test specified in 4.5.2.

3.5.3 Screwed. - The bolts shall have heads screwed onto the shanks. The heads, through which 3/8-inch 16NC2 or 16UNC2B holes have been tapped, shall be screwed tightly onto shanks, the upper ends of which have been threaded for a distance of approximately 7/16 inch with 3/8-inch 16NC3 or 16UNC2A threads. The part of the shanks which protrudes through the head shall then be heated and peened over so that it is impossible to unscrew the head from the bolt (see figure 3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. - All bolts of the same type, class, and nominal length offered for inspection at one time shall constitute a lot for purposes of inspection and tests.

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4.3 Sampling. -

4.3.1 Sampling for visual and dimensional examination. - Sampling for visual and dimensional examination shall be conducted in accordance with MIL-STD-105 at inspection level S-1 and acceptable quality level = 1.5 percent defective.

4.3.2 Sampling for chemical analysis. -

4.3.2.1 When varifiable evidence has been furnished by the manufacturer that the bar and rod stock used conforms to class A or B (see 3.1), as specified, two bolts for chemical analysis shall be taken from each lot to verify their nickel and chromium content.

4.3.2.2 When finished bolts cannot be identified as being processed from bar and rod stock conforming to class A or B (see 3.1) as specified, a sample of bolts shall be selected at random in accordance with table II for chemical analysis.

Table II - Sampling for chemical analysis and for tension load test of welded bolts.

Number of bolts in inspection lot	Number of bolts in sample
280 or less	3
281 - 1200	5
1201 - 10,000	8
10,001 or over	13

4.3.2.3 Representative millings or chips shall be taken from the head and from the shank of each sample bolt (from the head or the shank of each bolt in the case of forged bolts) and kept separate. In the case of welded bolts, samples for chemical analysis may be taken from tension load test specimens after tension testing. The samples shall be free from oil, dirt, grit or other foreign matter and shall consist of not less than 2 ounces.

4.3.3 Sampling for tension load tests (welded bolts only). - A random sample of welded bolts for tension load test shall be selected in accordance with table II.

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4.4 Visual and dimensional examination. -

4.4.1 Procedure. - Each of the sample bolts selected in accordance with 4.3.1 shall be examined for defects in appearance and dimensions as follows:

Examine	Defects
Appearance	Not type specified.
	Not shaped as shown on figure 1 or figure 2, as specified.
	Not clean, free of chips or dirt.
	Bolt head not fully formed (forged bolts).
	Head and shank not tapped and threaded as specified (screwed bolts).
	Protruding part of shank not peened over (screwed bolts).
Dimensions	Dimensions not within tolerances specified (see 3.4).

4.4.2 Acceptance criteria. - Any bolt in the sample containing one or more visual or dimensional defects shall be cause for rejection, and if the number of defective bolts in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, this shall be cause for rejection of the lot represented by the sample. Rejected lots may again be offered for inspection provided the contractor has reexamined each item of the lot and removed or reworked nonconforming bolts, and states that he has examined the entire lot and found it to conform with the requirements of this specification.

4.5 Tests. -

4.5.1 Chemical analysis. -

4.5.1.1 Procedure (applicable to 4.3.2.1 samples). - The samples obtained as specified in 4.3.2.1 shall be tested for nickel and chromium content in accordance with method 111.1 of FED-STD-151.

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4.5.1.1.1 Acceptance criteria. - If any sample fails to conform to the nickel or chromium content specified in 3.1 for class A or B bolts, as applicable, this shall be cause for rejection of the lot from which the samples were taken.

4.5.1.2 Procedure (applicable to 4.3.2.2 samples). - All samples selected in accordance with 4.3.2.2 shall be tested for content of nickel and chromium in accordance with method 111.1 of FED-STD-151. Two of the samples selected in accordance with 4.3.2.2 shall be tested for specified elements other than nickel and chromium in accordance with method 111.1 of FED-STD-151.

4.5.1.2.1 Acceptance criteria. - If any analysis result indicates nonconformance with a requirement of 3.1 for the class specified, this shall be cause for rejection of the lot from which the samples were taken.

4.5.2 Tension load test (welded bolts). - Each of the sample welded bolts selected in accordance with 4.3.3 shall be subjected to a tension load of 5000 pounds. The load shall be applied in such a manner as to subject the welded joint to the tension load.

4.5.2.1 Acceptance criteria. - If any bolt fails to withstand the tension load, this shall be cause for rejection of the lot represented by the sample.

5. PREPARATION FOR DELIVERY

5.1 Preparation for packaging and packing. - The bolts shall be clean, free of dirt, chips, or any foreign material. Contact preservatives shall not be used.

5.2 Packaging and packing. - The level of packaging and packing shall be as specified (see 6.1).

5.2.1 Level A. - When level A is specified, bolts shall be prepared for shipment in accordance with MIL-H-3982.

5.2.2 Level C. - When level C is specified, bolts shall be prepared for shipment in accordance with commercial practice to insure carrier acceptance and safe delivery at destination; and shall meet, as a minimum, the requirements of carrier rules and regulations applicable to the mode of transportation.

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5.3 Marking. - In addition to any special marking required by the contract or order (see 6.1), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Length of shank required, and if type b, the length of the hook end (see 3.2 and 3.3).
(d) Whether head of bolt should be other than specified (see 3.5).
(e) Selection of applicable level of packaging and packing (see 5.1 and 5.2).
(f) Additional marking, if required (see 5.3).

6.2 CHANGES FROM PREVIOUS ISSUE. - THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5306-N015Sh)

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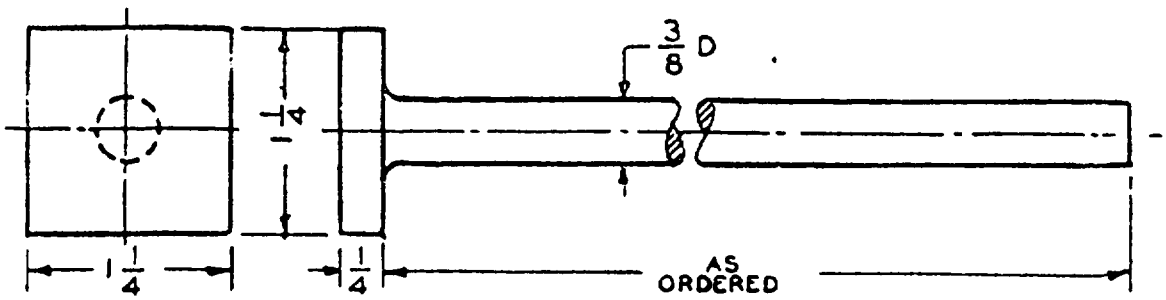


Figure 1 - Type a, straight end.

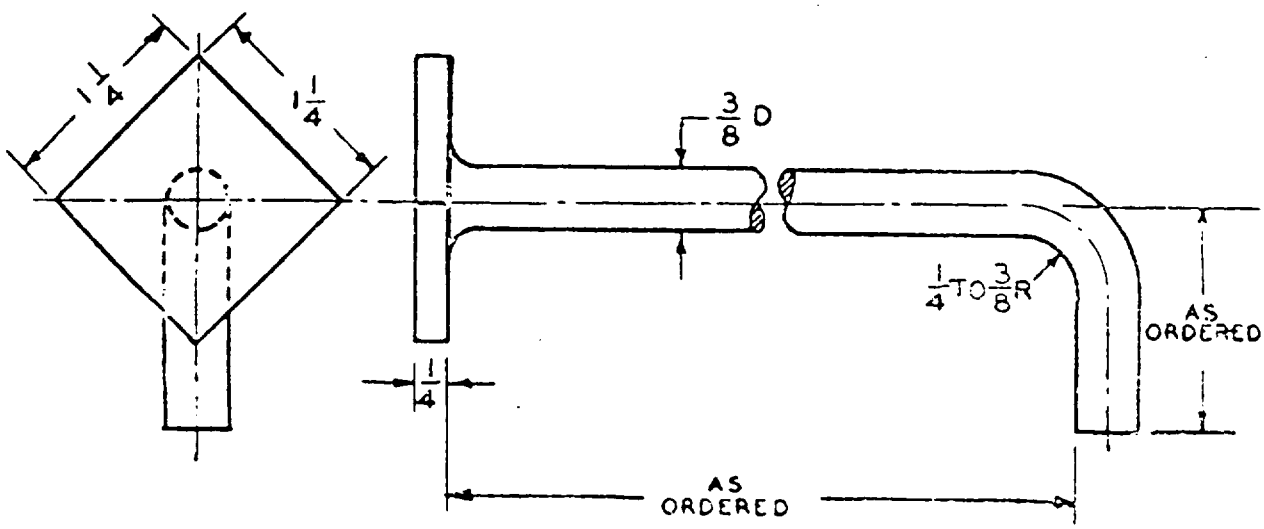


Figure 2 - Type b, hook end.

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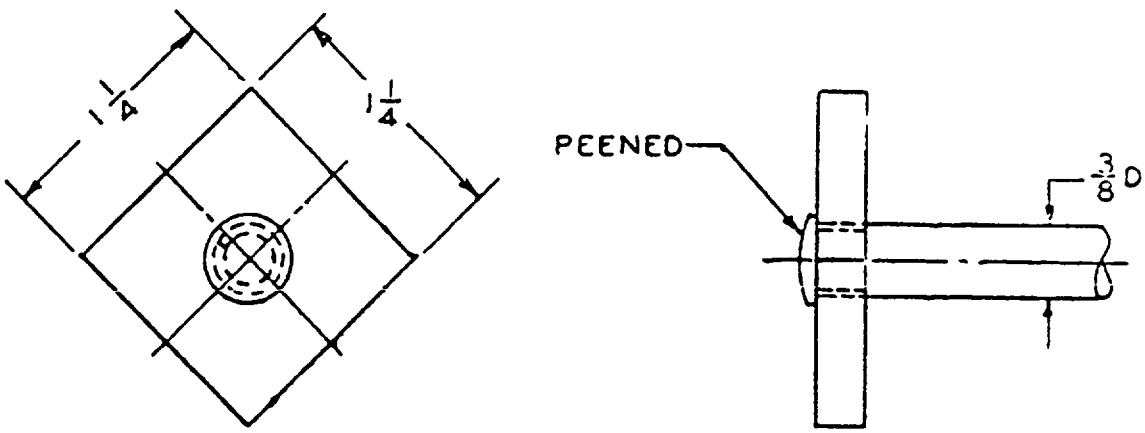


Figure 3 - Screwed and peened heads for type a or b anchor bolts.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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